

RAG Performance Analysis: Hybrid Retrieval Delivers Superior Results

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Comprehensive evaluation of 50 QA cases on Chroma vector database reveals significant performance gains with hybrid retrieval architecture



Default Retriever

Cosine Similarity: 0.45 max / 0.17 mean

ROUGE-1: 0.61 max / 0.20 mean

ROUGE-2: 0.53 max / 0.08 mean



Hybrid Retriever

Cosine Similarity: 0.65 max / 0.20 mean

ROUGE-1: 0.68 max / 0.23 mean

ROUGE-2: 0.67 max / 0.12 mean

Key Performance Gains

- **+44%** improvement in max cosine similarity
- **+15%** increase in mean similarity scores
- **+12%** boost in ROUGE-1 mean accuracy
- **+46%** enhancement in ROUGE-2 mean precision

Strategic Implication

Hybrid retrieval (vector + BM25) consistently outperforms default vector-only approaches across all semantic similarity and linguistic quality metrics, indicating improved answer relevance and precision for enterprise knowledge retrieval systems.